

Thermodynamic State Vector & Monadic Execution Framework in Web of Life: Sprint 012

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Current Sprint 2025

Abstract

We report the formal architectural completion and verification of Sprint 012, establishing the Thermodynamic State Vector (`src/thermodynamics/types.ts`) and monadic execution pipelines (`src/thermodynamics/thermodynamic.structure.ts`) within the Web of Life planetary simulation engine. Grounded in classical nonequilibrium thermodynamics and systems ecology, this sprint operationalizes strict mathematical contracts for internal entropy generation ($\dot{S}_{\text{gen}}$), exergy destruction rate ($\dot{I} = T_0 \dot{S}_{\text{gen}}$), and boundary flux arrays. By wrapping biogeochemical stocks and thermal transitions in a **ThermodynamicMonad**, the simulation mathematically enforces the First Law of Thermodynamics (energy conservation) and the Second Law of Thermodynamics (Clausius inequality, $\dot{S}_{\text{gen}} \geq 0$). This preprint outlines the theoretical framework, interface specifications, monad mechanics, and verification strategies deployed across simulated terrestrial pods. Official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.

1 Introduction and Systems Ecology Context

Simulating complex planetary ecosystems requires rigorous accounting not only of mass conservation across biogeochemical cycles (Carbon, Nitrogen, Phosphorus, and Water) but also of the thermodynamic driving forces that power ecological organization. According to nonequilibrium thermodynamics and systems ecology, living systems are open thermodynamic structures that maintain internal organization by exporting entropy to their environment while continuously dissipating exergy.

Sprint 012 bridges macroscopic thermodynamic principles with discrete-time simulation architectures. By embedding thermodynamic state tracking directly into the type system and execution pipeline, the Web of Life engine ensures that no ecological state transition violates physical laws.

2 Theoretical Framework & Mathematical Contracts

2.1 First Law of Thermodynamics (Energy Conservation)

The total internal energy change dU/dt of any planetary subsystem or control volume V is governed by:

$$\frac{dU}{dt} = \sum_i \dot{Q}_i - \dot{W}_{\text{sys}} + \sum_k \dot{m}_k \left(h_k + \frac{v_k^2}{2} + gz_k \right) \quad (1)$$

In our terrestrial pod context, mechanical work $\dot{W}_{\text{sys}}$ is negligible compared to thermal radiation fluxes and material enthalpy transport.

2.2 Second Law of Thermodynamics (Entropy Balance)

The entropy rate balance for an open thermodynamic system is defined as:

$$\frac{dS_{\text{sys}}}{dt} = \sum_j \frac{\dot{Q}_j}{T_j} + \sum_k \dot{m}_k s_k + \dot{S}_{\text{gen}} \quad (2)$$

Where $\dot{S}_{\text{gen}}$ is the internal entropy generation rate, satisfying the Clausius inequality:

$$\dot{S}_{\text{gen}} \geq 0 \quad (3)$$

2.3 Exergy Destruction Rate ($\dot{I}$)

Evaluated at the dead-state reference temperature $T_0 = 288.15$ K:

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \geq 0 \quad (4)$$

3 Monadic State Architecture

To guarantee invariant preservation during state propagation, transformations are encapsulated within `ThermodynamicMonad<T>`. Each bind operation evaluates:

$$\vec{V}_{t+\Delta t} = \vec{V}_t + \left(\frac{d\vec{V}}{dt} \right) \Delta t \quad (5)$$

If $\dot{S}_{\text{gen}} < 0$, a `SecondLawViolation` exception is thrown immediately.

4 Conclusion

Sprint 012 successfully establishes the rigorous thermodynamic backbone of the Web of Life engine. Source code, test suites, and documentation are available at <https://github.com/pascalranoroarijaona/WebOfLife>.